# Baihan Zhang

Stanford, CA | zhang.baihan02@gmail.com | github.com/zhangbaihan

## Education

**Stanford University** Stanford, CA
*M.S. in Computer Science* *Exp. Graduation Spring 2027*

- **Knowledge Domains:** VLMs/World Models, Reinforcement Learning, Sparse DNNs (MoE), Robotics, Interpretability, LLM Security (Prompt Injections, Jailbreaks), Linear Algebra.
- **Activities:** Columnist at Stanford Blockchain Review, Member at Stanford AI Alignment

## Experience

**Stanford NeuroAI Lab, advised by Prof. Dan Yamins** Stanford, CA
*Research Assistant (Building a World Model with Multimodal CoT)* *Feb 2026 – Present*

- Planned, executed, and evaluated 1.3M steps of pre-training for an autoregressive world model (PSI2) on 29 multimodal datasets.
- Created Qwen3-VL based pipeline for captioning egocentric and robotic videos to support PSI2 training - generated 2.5M examples.
- Training PSI2 to intelligently determine optimal sequence of intermediate modalities during Chain-of-Thought through reinforcement learning, alongside interpretability studies to quantify multimodal CoT's performance gain on difficult visual reasoning tasks.

**Stanford Computer Security Group, advised by Prof. Dan Boneh** Stanford, CA
*Research Assistant (Building Defense Against Agentic Prompt Injections)* *March 2026 – Present*

- First-authoring publication on novel prompt injection defensive system trained simultaneously on agent internal activation through SAE, and raw untrusted text that agent interacts with.
- Building a novel benchmark of 100k examples with 100% Attack Success Rate on LlaMa3-8B, designed to prevent defense probers from learning instruction and Regex-level shortcuts, using reinforcement learning red-teaming techniques.

**FANUC (Leading Industrial Robotic Arm Manufacturer)** Detroit, MI
*Product Management Intern* *March 2025 – June 2025*

- Spearheaded 90+ customer interviews and market research that reshaped corporate strategy, secured approval, and launched a seed financing program providing FANUC co-bots to pre-seed/Series A robotics startups.
- Assisted senior leaders with the assessment of project costs and usage scenarios, conducting feasibility analysis to validate business use cases.
- Served as a liaison between engineering and business teams, ensuring technical specifications aligned with customer scenarios.

**Stanford University, Kundaje Lab** Stanford, CA
*Undergraduate Research Assistant (Interpretability)* *Jan 2025 – March 2025*

- Used TF-MODISCo and DeepSHAP to reveal genomic model chromBPNet decision boundaries on high-dimensional thyroid cancer data.
- Reverse-engineered deep learning decision processes to assign feature contribution scores, identifying 2 novel proteins correlating to thyroid cancer occurrence.

## Projects

**Human-Behavior Predictor** | *Python* February 2026

- Developed screen monitoring tool on personal laptops
- Built agentic pipeline to synthesize user information into actionable memory of user habit, personality
- Fine-tuned LlaMa with LoRA to use synthesized user info to make prediction about user behavior on Spotify, Tinder, and Amazon.

**Sparse Neural Network Architecture Research** | *Python, PyTorch, COMET* Nov 2025

- Developed a diagnostic framework for Conditionally-Overlapping Mixture-of-Experts (COMET) models to analyze routing stability, routing collapse, and computational efficiency.

- Formulated novel "Mask Coherence" and "Feature Sensitivity" metrics to quantify semantic fragmentation and systematic robustness failure modes.
- Demonstrated via mathematical analysis that background noise, rather than semantic content, drove routing decisions in CIFAR-10 classifiers, exposing critical robustness failures in sparse architectures.

**TidyBot2: Speech-Controlled Robot** | *Python, ROS2, MuJoCo, Mink* Feb 2026

- Implemented a motion planner with joint limits, collision checking, and singularity detection, exposing ROS2 services for trajectory execution.
- Demoed with a mobile bimanual robot (3-DOF base, $2\times6$-DOF WX250s arms, grippers, pan-tilt RealSense) to recognize human command and fetch objects, performing reasoning to sort objects.

**Website Optimizer for On-Chain Agents (YC Hackathon Top 10)** | *React, AI Agents* Nov 2025

- Built agentic infrastructure converting e-commerce sites into structured formats (commerce.txt) for autonomous agent access.
- Designed a standardized data format to structure e-commerce information, establishing a clear schema for machine interaction and payment execution.
- Placed Top 10 out of 150 teams, selected to pitch directly to Y Combinator and Stripe partners.

**Quantum Circuit Analysis (BlueQubit Hackathon Winner)** | *Python, Qiskit, OpenQASM* Nov 2025

- Secured prize ($150 BTC) by solving the "One Peak" challenge: identifying max-amplitude bitstrings in increasingly complex OpenQASM 2.0 circuits.
- Engineered a Qiskit analysis script to parse qasm2 files and perform state vector simulations, effectively isolating the peak probability outcome.
- Optimized the search space for "peak bitstrings" by analyzing circuit amplitude distributions, effectively reverse-engineering the circuit logic.

**Ethereum Splitwise & Multisig** | *Solidity, Foundry, Graph Theory* Oct 2025

- Developed gas-optimized smart contracts implementing on-chain debt graphs with automatic cycle resolution algorithms.
- Verified system correctness and security behaviors through extensive Foundry unit tests, ensuring the immutability and safety of fund management logic.

**FilmR: CLIP-Based Video Retrieval System** | *PyTorch, CLIP, Computer Vision* March 2025 – June 2025

- Benchmarked a video retrieval system against baseline CLIP models under severe compute and data constraints.
- Achieved a 2.6% average gain in embedding diversity while maintaining object detection accuracy across 40+ complex queries.

**Plant Health Detector** | *TensorFlow, CNNs, Computer Vision* Jan 2025 – March 2025

- Designed a custom weighted CNN to detect diseased strawberry leaves, achieving 98.8% accuracy and outperforming Kaggle baselines by 4%.
- Curated a novel dataset by collecting and hand-labeling 800+ real-world farm images to improve model generalization.

## Technical Skills

**Languages**: C, C++, Python (PyTorch), JavaScript, HTML/CSS, Solidity
**Systems & Tools**: Linux, Git, MongoDB, AWS, LaTeX, Figma